AIM-Fair: Advancing Algorithmic Fairness via Selectively Fine-Tuning Biased Models with Contextual Synthetic Data

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Abstract

Models trained on real-world datasets often inherit demographic biases, leading to disparate performance across protected groups. Existing fairness approaches that leverage generative data struggle with low image quality, domain gaps between synthetic and real distributions, and reliance on costly demographic annotations. We propose ***AIM-Fair****, a novel framework that combines* Contextual Synthetic Data Generation *(CSDG) with* Selective Fine-Tuning *(SFT) to improve worst-group performance without sacrificing overall accuracy. CSDG employs GPT-4 to construct rich, context-aware prompts that guide a Stable Diffusion model to produce diverse, balanced synthetic images without requiring demographic annotations on real data. A* Selective Mask Generation *(SMG) module then identifies parameters that are simultaneously fairness-sensitive and domain-insensitive via gradient-difference ranking across three data streams: biased real data, biased synthetic data, and balanced synthetic data. Only the intersection of top-k fairness-sensitive and domain-insensitive parameters is updated during fine-tuning on the balanced synthetic data. Extensive experiments on CelebA and UTKFace demonstrate that AIM-Fair achieves state-of-the-art fairness–utility trade-offs, outperforming prior generative methods and standard fine-tuning baselines across multiple attribute settings. Notably, on CelebA (Smiling/Male), AIM-Fair attains 89.02% accuracy, 84.20% worst-group accuracy, and 6.07% equalized odds, compared to 88.40%, 81.90%, and 7.40% for the strongest baseline DiGA.*

1. Introduction

Facial attribute classification (FAC) is a widely studied computer vision task with applications in identity verification, human–computer interaction, and content moderation [10, 24]. However, models trained with standard empirical risk minimization (ERM) on real-world datasets can inherit and amplify demographic biases present in the data [26, 41]. For example, a "Smiling" classifier trained on CelebA may learn to exploit gender as a spurious cue because the dataset contains a higher proportion of smiling female faces, causing significantly worse performance for minority demographic groups.

Addressing such *algorithmic unfairness* has motivated a rich body of work. Reweighting and distributionally robust optimization (DRO) methods [9, 12, 21] aim to equalize performance across groups but typically require group annotations at training time. Invariant learning approaches [1, 5] seek features that generalize across environments but often require explicit environment labels. Fairness-constrained learning [8, 11, 25] directly penalizes disparities but may struggle when demographic labels are unavailable.

A promising alternative is to leverage *generative data augmentation*: synthesizing balanced training examples to counteract the bias in real data [28, 36, 43, 44]. Methods such as DiGA [44] use generative models to edit spurious attributes in existing images and train the classifier to be invariant to these edits. While effective, such approaches face three core challenges: (1) generated images often suffer from domain gaps with respect to real training data; (2) prompt design for diverse, attribute-balanced generation is non-trivial; and (3) naive fine-tuning on synthetic data risks degrading the feature representations acquired during pretraining on real data.

We present **AIM-Fair** (**A**lgorithm **I**mprovement for **M**odel **Fair**ness), a two-stage framework illustrated in Fig. 2 that addresses these challenges. Our key insight is that high-quality, contextually diverse synthetic data combined with *selective* parameter fine-tuning can substantially improve worst-group performance without compromising overall accuracy. The method operates as follows:

1. **Contextual Synthetic Data Generation (CSDG).** We query GPT-4 to generate diverse, descriptive prompts encompassing a wide range of facial attributes (skin tone, hair style, head pose, accessories, *etc.*), then use Stable Diffusion [33] to generate large balanced synthetic datasets conditioned on these prompts. This produces high-quality images covering under-represented demographic combinations.
2. **Selective Mask Generation (SMG).** Starting from a pretrained biased model, we compute parameter-wise gradients on three data streams: biased real data (g_R), bi-

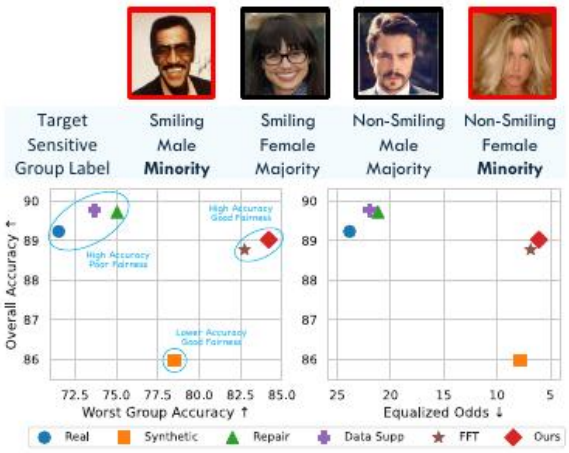

Figure 1. Facial attributes classification (FAC) on the CelebA dataset based on different training strategies, where ***Smiling*** is the target attribute and ***Gender*** is the protected attribute. AIM-Fair achieves a superior balance between overall accuracy and worst-group accuracy compared to ERM, full fine-tuning, and training purely on synthetic data.

ased synthetic data (g_{S1}), and balanced (unbiased) synthetic data (g_{S2}). We rank parameters by their *fairness sensitivity* $|g_{S1} - g_{S2}|$ and their *domain insensitivity* $|g_R - g_{S1}|$, then select the top-k parameters common to both rankings.

3. **Selective Fine-Tuning (SFT).** Only the selected parameters are updated while training on the balanced synthetic data, preserving domain-specific representations from pretraining while encouraging fairness improvements.

Our main contributions are:

- We propose AIM-Fair, a fine-tuning framework that improves model fairness using synthetic data without requiring demographic annotations on real data.
- We introduce CSDG, which uses GPT-4 generated contextual prompts with Stable Diffusion to generate diverse, high-quality, attribute-balanced images.
- We develop SMG, a gradient-difference based parameter selection mechanism that identifies fairness-sensitive and domain-insensitive parameters for selective fine-tuning.
- AIM-Fair achieves state-of-the-art fairness–utility trade-offs on CelebA and UTKFace, outperforming full/partial fine-tuning and prior generative methods across multiple target and protected attribute settings.

2. Related Work

2.1. Fairness in Machine Learning

Algorithmic fairness has been studied extensively in the machine learning literature [2, 4, 26, 38]. Common fairness metrics include equalized odds [11], which requires equal true and false positive rates across groups, and worst-group accuracy (WGA), which measures the minimum accuracy over all demographic subgroups [12]. Methods for improving fairness can be broadly categorized into pre-processing (data rebalancing, reweighting), in-processing (constrained optimization, adversarial training), and post-processing (threshold calibration) approaches [30].

DRO-based approaches [9, 12, 21] minimize the worst-case loss over distributional perturbations, providing robustness across subpopulations without explicit group labels. Methods like JTT [22] and LfF [27] identify and upweight misclassified examples likely to belong to minority groups. EIIL [5] partitions data into environments to learn invariant features. MAPLE [46] combines model patching with environment inference. Fair contrastive learning [28] disentangles target and sensitive attributes in the representation space.

In facial analysis specifically [6, 10, 16, 41], bias mitigation has been explored through balanced sampling, adversarial debiasing [25, 39], and causal intervention [17, 23]. Our method differs from these works by not requiring group-annotated real data.

2.2. Synthetic Data for Fairness

Generative approaches to fairness augment or replace biased training data with synthetically generated balanced examples. GAN-based methods [3, 29, 31] synthesize minority-group samples to rebalance training sets, but often produce low-quality images and lack diversity. More recent works leverage diffusion models for their superior image quality [33, 34]. ITI-Gen [43] learns prompt embeddings from reference images to generate inclusive images. FairRAG [36] uses retrieval-augmented generation for fair human synthesis. DiGA [44], the closest work to ours, uses generative models to edit spurious attributes and trains a model invariant to these edits. However, DiGA performs edits in image space and relies on identifying spurious attributes explicitly. Our CSDG approach instead generates entirely new diverse images through GPT-4 prompting, achieving better distribution coverage.

Data supplementation with synthetic images [35, 37, 40, 47] has been explored for general long-tail and data-scarce settings, but typically does not address the selective parameter update challenge that is central to AIM-Fair.

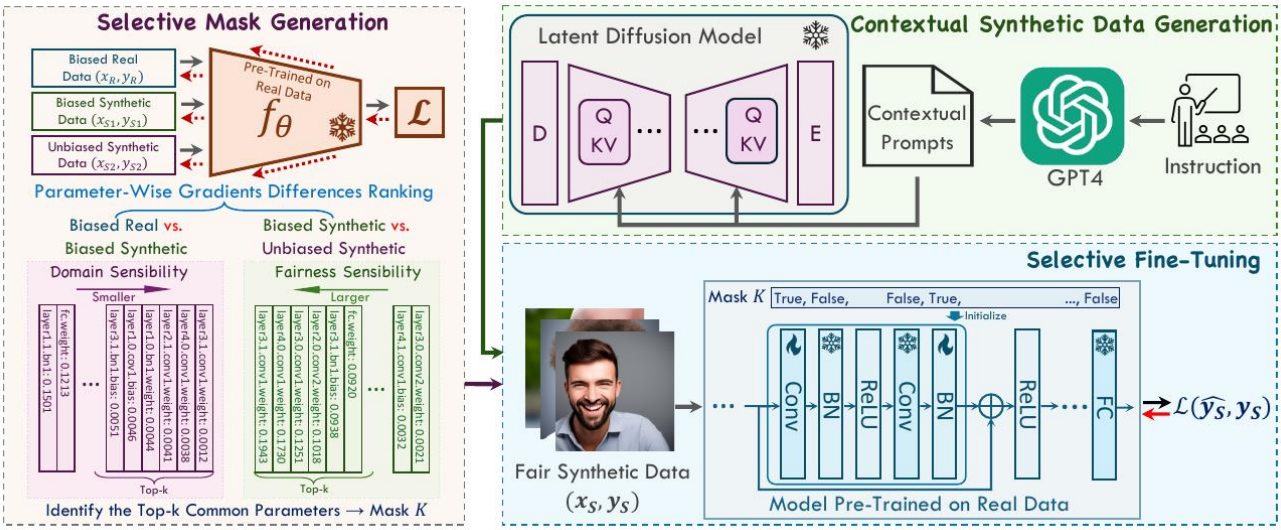

Figure 2. Overview of the AIM-Fair framework. It consists of three modules: (1) **CSDG**: Contextual Synthetic Data Generation using GPT-4 prompts to guide Stable Diffusion; (2) **SMG**: Selective Mask Generation that computes gradient differences to identify fairness-sensitive and domain-insensitive parameters; (3) **SFT**: Selective Fine-Tuning that updates only the identified parameters on balanced synthetic data to improve fairness while preserving domain utility.

2.3. Selective and Parameter-Efficient Fine-Tuning

Transfer learning and fine-tuning are fundamental tools in computer vision [14, 19, 48]. Parameter-efficient fine-tuning methods such as LoRA [15] and adapter tuning selectively update a small fraction of parameters. Sub-tuning approaches [18] freeze or update specific network blocks to preserve pretrained representations. Surgical fine-tuning [20] identifies which layers to fine-tune based on task-specific criteria. AutoLR [32] assigns layer-specific learning rates. Our SMG is complementary to these approaches: rather than using fixed architectural criteria, we select parameters based on their gradient responses to fairness-relevant and domain-relevant data, enabling a more targeted improvement. Domain gap embeddings [40] address the synthetic-to-real transfer problem at the representation level, which is orthogonal to our parameter selection strategy.

3. Method

3.1. Problem Formulation

Let $\mathcal{D}_R = \{(x_i, y_i, s_i)\}_{i=1}^N$ be a biased real training dataset where x_i is an image, $y_i \in \{0, 1\}$ is the target attribute (e.g., Smiling), and $s_i \in \{0, 1\}$ is a protected attribute (e.g., Male). The dataset is biased in the sense that y and s are spuriously correlated—the label distribution of y differs significantly between groups defined by s. We seek to train a classifier $f_\theta : \mathcal{X} \to \{0, 1\}$ that achieves both high overall accuracy and fairness, measured by worst-group accuracy (WST) and equalized odds (EO).

Crucially, we assume that real-data demographic annotations $\{s_i\}$ are *not* used during training (only target labels $\{y_i\}$ are available), making our setting practically relevant for scenarios where protected attribute labeling is expensive or privacy-sensitive.

3.2. Contextual Synthetic Data Generation (CSDG)

A key challenge in using generative models for fairness is producing diverse images that cover all demographic combinations of target and protected attributes. Naive prompts (e.g., "a photo of a smiling male") produce low-diversity outputs concentrated on common archetypes.

GPT-4 Prompt Generation. We query GPT-4 to generate a large set of diverse, contextually rich text prompts for each combination $(y, s) \in \{0, 1\}^2$. Prompts describe fine-grained facial attributes such as skin tone, hair color and style, age range, head pose, accessories (glasses, hats), makeup, lighting conditions, and background. For example, for the $(y = 1, s = 1)$ (Smiling Male) class, prompts include descriptions such as "a photo of a middle-aged man with a warm smile, salt-and-pepper beard, wearing rimless glasses, outdoor lighting."

Stable Diffusion Generation. Each GPT-4 generated prompt is used to condition Stable Diffusion [33] to generate multiple images. By sampling from a large pool of diverse prompts (≥ 50 per class combination), CSDG produces a *balanced synthetic dataset* $\mathcal{D}_{S2}$ with roughly equal representation across all (y, s) groups. We also generate

Figure 3. Generated images of *Smiling Male* conditioned on different GPT-4 prompts by CSDG. The diversity of facial attributes (skin tone, hair style, age, accessories, head pose) demonstrates the broad coverage of contextual prompts.

a *biased synthetic dataset* $\mathcal{D}_{S1}$ that mirrors the bias ratio of the real data (by sampling predominantly from majority-group prompts).

Figure 3 illustrates generated images for the *Smiling Male* class conditioned on different GPT-4 prompts, demonstrating the high diversity achieved by CSDG.

3.3. Selective Mask Generation (SMG)

Given a model f_θ pretrained on biased real data $\mathcal{D}_R$, we seek to identify which parameters to update during fair fine-tuning. Two desiderata guide our selection:

1. **Fairness sensitivity:** Parameters whose gradients differ significantly between biased and unbiased data are likely responsible for encoding bias-related features and are good candidates for update.
2. **Domain insensitivity:** Parameters whose gradients are similar between real and synthetic data are less likely to encode domain-specific features (e.g., image artifacts specific to real photography). Updating these parameters is safer from a domain-shift perspective.

Gradient computation. For each of the three datasets $\mathcal{D}_R$, $\mathcal{D}_{S1}$, and $\mathcal{D}_{S2}$, we compute parameter-wise aggregated gradients:

$$g_R = \nabla_\theta \mathcal{L}(\mathcal{D}_R), \quad g_{S1} = \nabla_\theta \mathcal{L}(\mathcal{D}_{S1}), \quad g_{S2} = \nabla_\theta \mathcal{L}(\mathcal{D}_{S2}), \tag{1}$$

where $\mathcal{L}$ is the training loss (binary cross-entropy).

Algorithm 1 AIM-Fair

Require: Biased real data $\mathcal{D}_R$; pretrained model f_θ; top-k budget

[CSDG] Generate prompts via GPT-4 for all (y, s) groups
  Generate balanced synthetic data $\mathcal{D}_{S2}$ via Stable Diffusion
  Generate biased synthetic data $\mathcal{D}_{S1}$ (mirrors $\mathcal{D}_R$ bias)
[SMG] Compute g_R, g_{S1}, g_{S2} on pretrained f_θ
  Rank by fairness sensitivity $\Delta_{\text{fair}} = |g_{S1} - g_{S2}|$
  Rank by domain sensitivity $\Delta_{\text{domain}} = |g_R - g_{S1}|$
  Compute mask $\mathcal{M} = \mathcal{T}_{\text{fair}} \cap \mathcal{T}_{\text{domain}}$
[SFT] Freeze $\theta_j \notin \mathcal{M}$; fine-tune $\theta_j \in \mathcal{M}$ on $\mathcal{D}_{S2}$
return Fair model f_θ

Ranking. For each parameter θ_j, we compute:

$$\Delta^{(j)}_{\text{fair}} = |g^{(j)}_{S1} - g^{(j)}_{S2}|, \quad \text{(fairness sensitivity)} \tag{2}$$

$$\Delta^{(j)}_{\text{domain}} = |g^{(j)}_{R} - g^{(j)}_{S1}|, \quad \text{(domain sensitivity)} \tag{3}$$

where superscript (j) denotes the mean absolute value over entries of the parameter tensor.

Selective mask. Let $\mathcal{T}_{\text{fair}}$ denote the top-k parameters ranked by Δ_{fair} in *descending* order (highest fairness sensitivity), and $\mathcal{T}_{\text{domain}}$ the top-k parameters ranked by Δ_{domain} in *ascending* order (lowest domain sensitivity). The final update mask is:

$$\mathcal{M} = \mathcal{T}_{\text{fair}} \cap \mathcal{T}_{\text{domain}}. \tag{4}$$

This intersection selects parameters that can be modified to improve fairness while being safe to update without disrupting real-domain representations.

3.4. Selective Fine-Tuning (SFT)

Using the mask $\mathcal{M}$, we freeze all parameters $\theta_j \notin \mathcal{M}$ and fine-tune only $\theta_j \in \mathcal{M}$ on the balanced synthetic dataset $\mathcal{D}_{S2}$ using SGD:

$$\theta_j \leftarrow \theta_j - \eta \nabla_{\theta_j} \mathcal{L}(\mathcal{D}_{S2}), \quad \forall j \in \mathcal{M}, \tag{5}$$

with learning rate η and standard momentum and weight decay. The full algorithm is summarized in Algorithm 1.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate on two standard facial attribute benchmarks:

- **CelebA** [24]: 202,599 celebrity face images with 40 binary attribute annotations. We consider two target attributes (Smiling, Young) and two protected attributes (Male, Young), giving three evaluation settings.

- **UTKFace** [45]: 20,000+ face images with age, gender, and race annotations. We use Female as the target and White as the protected attribute.

Metrics. Following prior work [28, 44], we report:

- **ACC**: Overall classification accuracy (%, ↑).
- **WST**: Worst-group accuracy (%, ↑)—minimum accuracy over all (y, s) subgroups.
- **EO**: Equalized odds (%, ↓)—difference in true positive rates across protected groups.
- **STD**: Standard deviation of group accuracies (%, ↓).

Implementation. We use ResNet-50 [13] pretrained on ImageNet as the backbone. Pre-training uses SGD with lr=0.01, momentum=0.9, weight decay=10^{-4}, for 15 epochs on biased real data. Fine-tuning uses lr=0.05 for 10 epochs on the balanced synthetic data. We set $k = 40$ for the top-k selection (ablated in Sec. 4.5). Images are resized to 224×224. CSDG generates 5,000 images per (y, s) group for balanced data, and biased synthetic data uses a 1:9 bias ratio (ablated in Sec. 4.5). All results are averaged over multiple random seeds.

Baselines. We compare against: ERM (ResNet-50 [13]), CVaR DRO [21], EIIL [5], LfF [27], JTT [22], MAPLE [46], and DiGA [44].

4.2. Comparison with State-of-the-Art Methods

Table 1 presents comparisons to prior methods on CelebA under three attribute settings. AIM-Fair achieves the best or second-best performance on all three fairness metrics in all settings:

- **T: Smiling, P: Male**: AIM-Fair achieves ACC=89.02, WST=84.20, EO=6.07, outperforming DiGA (ACC=88.40, WST=81.90, EO=7.40) on all metrics.
- **T: Smiling, P: Young**: AIM-Fair achieves ACC=90.21, WST=87.78, EO=5.44, markedly better than DiGA (ACC=89.10, WST=78.50, EO=9.50), with a 9.28pp improvement in WST.
- **T: Young, P: Male**: AIM-Fair obtains WST=54.89 and EO=28.18, substantially improving over DiGA (WST=51.30, EO=33.30), though with slightly lower ACC (78.19 vs. 80.00).

These results confirm that AIM-Fair consistently achieves superior fairness–utility trade-offs. The only setting where AIM-Fair does not lead in ACC (T: Young, P: Male) is also the setting with the largest fairness improvement, suggesting that a small accuracy trade-off can sometimes enable larger fairness gains.

4.3. Comparison of Training Strategies

Table 3 compares different approaches to using synthetic data for training: training purely on synthetic data, data supplementation [35], data repairing [7, 42], linear probing, full fine-tuning, and AIM-Fair.

Training purely on synthetic data significantly reduces EO (e.g., from 23.84 to 7.87 for CelebA T:Smiling/P:Male) but comes with a 3+ point drop in ACC, highlighting the domain gap between synthetic and real distributions. Full fine-tuning on synthetic data achieves better accuracy while maintaining low EO (6.83), but AIM-Fair's selective approach further improves WST (84.20 vs. 82.74) and reduces EO (6.07 vs. 6.83). Data supplementation and repairing approaches maintain high accuracy but show much worse fairness (EO $>$ 20), indicating they do not effectively address the bias. Across all datasets and settings, AIM-Fair provides the best or second-best fairness metrics while maintaining competitive accuracy.

4.4. Data-Scarce Settings

Table 2 (embedded in Tab. 1) evaluates performance under reduced real training data. At 50% of data, AIM-Fair achieves ACC=88.85, WST=83.99, EO=6.28; at 25%, ACC=88.89, WST=82.54, EO=7.59; and at 10%, ACC=87.90, WST=81.73, EO=8.16. AIM-Fair consistently outperforms all baselines including DiGA at all data fractions. Notably, DiGA achieves best ACC at 10% data (88.30 vs. 87.90) but AIM-Fair leads in WST and EO at all scales. These results demonstrate that AIM-Fair is especially beneficial in data-scarce settings, where real demographic group coverage is limited.

4.5. Ablation Studies

Effect of Prompt Design. Table 4 analyzes the impact of prompt diversity on the quality of generated data. Using a single plain prompt produces severely degraded results (WST=43.28, EO=34.20 for T:Smiling/P:Male), confirming that prompt diversity is essential. Increasing contextual prompts from 25 to 50 improves WST but may not always reduce EO. Crucially, adding head pose variation to contextual prompts (50 contextual + head poses) achieves the best overall performance (ACC=85.98, WST=78.52, STD=4.07), underscoring the role of head pose diversity in generating comprehensive training data. When used as the synthetic data for the full AIM-Fair pipeline (with selective fine-tuning), this translates to the best fairness results.

Effect of Number of Synthetic Samples. Table 5 shows results for varying the ratio of synthetic to real data. Using a 1:1 ratio (synthetic:real) achieves the best WST (84.20) and EO (6.07), while using only half as many synthetic samples reduces WST by 3.4pp. Doubling the synthetic data to a 2:1 ratio slightly reduces performance (WST=83.36), possibly due to overfitting to the synthetic distribution. These results suggest that matching synthetic to real data volume is optimal.

Effect of Bias Ratio in Synthetic Data. Table 6 examines how the bias ratio of the biased synthetic data $\mathcal{D}_{S1}$ affects SMG. A stronger imbalance (1:9 bias ratio, meaning

Table 1. Comparisons to other methods on the CelebA dataset under settings of varied target and protected attributes.

Methods	T: Smiling, P: Male			T: Smiling, P: Young			T: Young, P: Male		
	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)
ERM [13]	88.20	70.10	25.30	88.30	71.50	15.60	77.70	42.00	52.00
CVaR DRO [21]	87.30	74.00	22.80	87.00	76.10	13.90	75.40	42.30	48.80
EIIL [5]	87.90	75.60	19.70	87.90	72.50	13.30	77.50	45.60	39.20
LfF [27]	87.10	77.50	17.00	85.30	72.90	14.30	77.40	44.20	43.60
JTT [22]	88.00	74.80	19.40	87.60	73.30	14.20	76.30	43.60	47.70
MAPLE [46]	88.10	72.00	19.60	88.10	73.60	13.60	76.30	46.20	43.50
DiGA [44]	88.40	81.90	7.40	89.10	78.50	9.50	**80.00**	51.30	33.30
AIM-Fair (Ours)	**89.02**	**84.20**	**6.07**	**90.21**	**87.78**	**5.44**	78.19	**54.89**	**28.18**

Table 2. Comparisons to other methods on the CelebA dataset (T=Smiling, P=Male) under settings of training set sizes.

Methods	Samples Ratio = 50%			Samples Ratio = 25%			Samples Ratio = 10%		
	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)	ACC (↑)	WST(↑)	EO (↓)
ERM [13]	87.50	67.80	26.10	87.10	65.90	27.70	86.90	62.80	28.90
CVaR DRO [21]	86.60	72.90	22.10	86.60	72.30	22.40	85.50	69.10	27.30
EIIL [5]	86.20	71.30	22.50	85.90	69.60	25.40	86.80	64.20	26.70
LfF [27]	86.90	75.50	19.40	85.90	72.10	23.60	85.50	66.10	27.70
JTT [22]	87.30	72.90	20.10	86.70	71.10	20.60	86.80	67.10	23.10
MAPLE [46]	87.40	73.70	23.80	87.00	72.70	24.20	85.60	69.20	27.10
DiGA [44]	88.40	81.10	7.80	88.40	78.30	8.00	**88.30**	78.80	8.40
AIM-Fair (Ours)	**88.85**	**83.99**	**6.28**	**88.89**	**82.54**	**7.59**	87.90	**81.73**	**8.16**

Table 3. Comparisons of varied training strategies on CelebA and UTKFace datasets.

Methods	Target	CelebA: Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young				UTKFace: Protected		T: Female; P: White			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Baseline [13]	T=0	83.57	98.50	89.23	71.52	23.84	10.65	78.72	96.09	90.19	78.72	17.37	6.92	79.58	96.16	88.86	79.58	16.59	7.26
	T=1	95.36	71.52					94.37	86.42					95.75	83.96				
Trained on Synthetic Data	T=0	86.82	89.40	85.98	78.52	7.87	4.07	85.45	85.76	85.12	82.42	**3.50**	**1.34**	82.71	88.12	85.44	82.71	5.50	**1.96**
	T=1	86.39	78.52					82.42	85.27					84.90	86.04				
Data Supplementation [35]	T=0	84.39	98.39	**89.77**	73.67	21.96	9.76	78.05	95.36	90.35	78.05	17.31	7.02	82.12	93.97	**90.25**	82.12	11.85	5.24
	T=1	95.40	73.67					94.90	87.62					95.62	89.28				
Data Repairing [7, 42]	T=0	82.82	98.10	89.72	75.05	21.18	9.50	79.50	95.19	**90.68**	79.50	15.69	6.30	81.15	94.22	89.42	81.15	13.37	5.89
	T=1	96.04	75.05					94.51	88.35					95.71	86.61				
Linear Prob	T=0	84.55	96.22	89.69	77.41	17.51	7.71	85.20	93.81	90.14	85.20	8.61	3.18	81.06	91.51	88.40	81.06	10.59	5.14
	T=1	94.82	77.41					90.43	87.88					94.59	86.42				
Fully Fine-Tuning	T=0	88.81	91.62	88.77	82.74	6.83	3.31	85.98	90.06	87.85	85.98	4.37	1.68	83.00	88.04	87.90	83.00	5.40	2.99
	T=1	89.55	82.74					86.02	86.81					90.74	89.83				
AIM-Fair (Ours)	T=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**	88.19	93.64	90.21	**87.78**	5.44	2.34	84.26	89.08	88.30	**84.26**	**4.81**	2.41
	T=1	90.25	84.20					88.98	87.78					90.62	89.25				

Table 4. Classification results on CelebA dataset (T=Smiling, P=Male) under settings of different prompt types and numbers.

Prompt Types (Number)	Targe	Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Plain Prompt (1)	t=0	65.18	89.44	71.65	43.28	34.20	17.07	83.10	77.72	72.75	59.53	9.30	8.96
	t=1	77.47	43.28					59.53	68.83				
Contextual Prompts (25)	t=0	77.12	89.25	81.85	69.73	16.3	7.65	76.72	67.90	79.80	67.90	8.82	8.75
	t=1	85.99	69.73					85.16	91.09				
Contextual Prompts (50)	t=0	89.47	89.80	82.66	72.22	**5.07**	7.66	87.72	90.43	**85.85**	79.67	4.36	4.20
	t=1	77.26	72.22					79.67	82.67				
Contextual + Head Poses (50)	t=0	86.82	89.40	**85.98**	**78.52**	7.87	**4.07**	85.45	85.76	85.12	**82.42**	**3.50**	**1.34**
	t=1	86.39	78.52					82.42	85.27				

Table 5. Classification results on CelebA dataset (T=Smiling, P=Male) with different number of synthetic data.

Ratio To Real Data	Target	Protected		ACC (↑)	WST (↑)	EOD (↓)	STD (↓)
		P=0	P=1				
0.5	t=0	88.65	93.96	**89.51**	80.80	10.31	4.90
	t=1	91.10	80.80				
1	t=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**
	t=1	90.25	84.20				
1.5	t=0	89.18	92.05	88.34	81.25	7.13	3.98
	t=1	88.38	81.25				
2	t=0	89.01	91.26	88.81	83.36	6.08	2.96
	t=1	91.26	89.44				

10% majority class) yields the best WST (84.20) and EO (6.07), outperforming more moderate ratios (4:6, 3:7, 2:8) and even full fine-tuning. This makes sense: a more extreme imbalance in $\mathcal{D}_{S1}$ creates a larger gradient difference between g_{S1} and g_{S2}, making fairness-sensitive parameters easier to identify.

Effect of Top-k Selection. Figure 4 shows classification results for different values of k in the SMG top-k selection. Performance peaks around $k = 40$ and degrades at very small k (too few parameters updated) and very large k

Table 6. Classification results on CelebA dataset (T=Smiling, P=Male) with varied bias ratio of the biased synthetic data.

Bias Ratio	Target	Protected		ACC (↑)	WST (↑)	EOD (↓)	STD (↓)
		P=0	P=1				
Fully Fine-Tuning	t=0	88.81	91.62	88.77	82.74	6.83	3.31
	t=1	89.55	82.74				
4 : 6	t=0	85.72	91.71	**89.23**	83.90	8.78	3.77
	t=1	92.69	83.90				
3 : 7	t=0	87.05	91.62	88.90	83.52	7.72	3.26
	t=1	90.93	83.52				
2 : 8	t=0	88.01	91.76	88.97	83.16	7.30	3.28
	t=1	90.46	83.16				
1 : 9	t=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**
	t=1	90.25	84.20				

(including domain-sensitive parameters, hurting accuracy). This confirms that there is a sweet spot in the number of parameters to update.

Comparison of Partial Fine-Tuning Strategies. Table 7 compares AIM-Fair's parameter selection against alternative partial fine-tuning strategies: random selection, Sub-Tuning [18] (update or freeze one network block), and cosine similarity-based selection. AIM-Fair's gradient-difference selection achieves the best WST and EO on CelebA (Smiling/Male), outperforming random selection (WST=82.17, EO=9.14), block-wise Sub-Tuning approaches, and cosine similarity selection (WST=83.33, EO=8.40). While Sub-Tuning by updating one block achieves slightly better overall ACC in some settings, it comes at the cost of significantly higher EO and lower WST, confirming that AIM-Fair's parameter-level selection is more targeted for fairness improvement.

Table 7. Comparisons of varied partial fine-tuning strategies on CelebA and UTKFace datasets.

Methods	Target	CelebA												UTKFace							
		Protected		T: Smiling; P: Male				Protected		T: Smiling; P: Young				Protected		T: Female; P: White			
		P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)	P=0	P=1	ACC (↑)	WST (↑)	EO (↓)	STD (↓)
Best Random Selection	T=0	87.93	92.81	89.33	82.17	9.14	4.09	85.52	91.53	88.95	85.52	6.02	2.16	83.50	88.85	87.92	83.50	5.38	2.61
	T=1	91.31	82.17					88.25	87.63					90.30	89.01				
Best Sub-Tuning [18] (By Updating One Block)	T=0	87.50	94.75	**90.04**	80.64	12.45	5.52	87.28	94.63	**90.33**	87.28	7.34	3.00	83.05	89.56	**89.01**	83.05	6.51	3.56
	T=1	93.09	80.64					89.15	87.32					92.07	91.34				
Best Sub-Tuning [18] (By Freezing One Block)	T=0	88.69	91.09	89.06	83.48	7.00	3.00	86.55	91.89	88.65	86.55	**5.34**	2.25	83.58	88.34	87.42	83.58	**4.76**	**2.23**
	T=1	90.48	83.48					87.08	86.55					88.96	88.81				
Selective Fine-Tuning (Cosine Similarity)	T=0	86.07	91.77	88.85	83.33	8.40	3.61	87.43	92.96	90.26	87.43	5.53	**2.05**	83.72	88.82	88.21	83.72	5.10	2.64
	T=1	91.53	83.33					89.52	88.65					90.14	90.15				
AIM-Fair (Ours) (Absolute Difference)	T=0	88.16	91.40	89.02	**84.20**	**6.07**	**2.74**	88.19	93.64	90.21	**87.78**	5.44	2.34	84.26	89.08	88.30	**84.26**	4.81	2.41
	T=1	90.25	84.20					88.98	87.78					90.62	89.25				

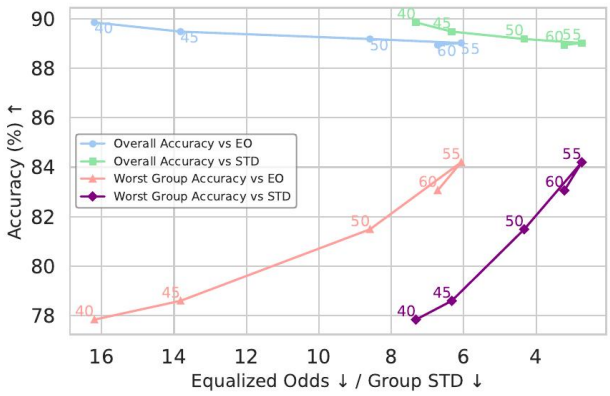

Figure 4. Classification results on CelebA (T=Smiling, P=Male) with different top-k values used in SMG. Performance peaks around $k = 40$, balancing sufficient parameter updates with preservation of domain-specific representations.

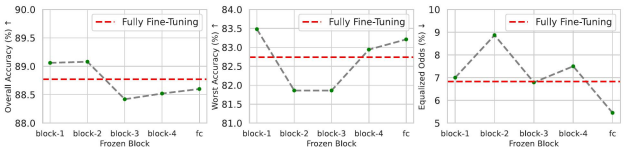

Figure 5. Effects of selective fine-tuning by layer-wise freezing on CelebA (T=Smiling, P=Male). Freezing only the FC layer improves WST and reduces EO compared to full fine-tuning, but AIM-Fair's parameter-wise selection consistently outperforms layer-wise strategies.

Layer-Wise Analysis. Figure 5 provides insight into which layers contribute most to fairness. The results show that simply freezing the fully connected (FC) layer already reduces bias compared to full fine-tuning, but does not achieve AIM-Fair's performance. Layers in the later ResNet blocks (layer3, layer4) show higher fairness sensitivity, consistent with the observation that later layers capture more task-specific and potentially biased features.

5. Discussion

Why does selective fine-tuning work? The intuition behind AIM-Fair is that bias in neural networks is encoded in specific parameters—those that strongly respond to the spurious correlation between target and protected attributes. By identifying these parameters via gradient differences between biased and unbiased data, and ensuring they are also insensitive to the domain gap (i.e., safe to update), we can surgically modify the model to reduce bias without disrupting its general visual representations.

Annotation-free operation. A key practical advantage of AIM-Fair is that it does not require demographic annotations on real data. The only labels needed are the target attribute annotations (e.g., Smiling=1/0), which are standard practice. The balanced synthetic data is constructed using the generation process itself, not from real-data group labels. This makes AIM-Fair directly applicable to settings where protected attribute labels are unavailable or cannot be collected due to privacy constraints.

Generalization across settings. AIM-Fair's strong performance across multiple attribute combinations (CelebA: Smiling/Male, Smiling/Young, Young/Male; UTKFace: Female/White) and data scales (10–100% of training data) demonstrates its broad applicability.

Limitations. AIM-Fair depends on the quality of the generative model: if Stable Diffusion fails to capture certain demographic groups accurately, the balanced dataset may still have residual coverage gaps. Additionally, the computational overhead of GPT-4 prompt generation and large-scale Stable Diffusion synthesis may be prohibitive for some applications. The gradient-based parameter selection requires three separate forward-backward passes over the datasets, though these can be parallelized.

6. Conclusion

We presented AIM-Fair, a two-stage framework for improving algorithmic fairness in facial attribute classification without requiring demographic annotations on real data. By combining Contextual Synthetic Data Generation (CSDG) using GPT-4 prompts with Stable Diffusion, and Selective

Fine-Tuning (SFT) guided by a gradient-based parameter selection module (SMG), AIM-Fair achieves state-of-the-art fairness–utility trade-offs on both CelebA and UTKFace. Ablation studies confirm the importance of prompt diversity, the bias ratio in synthetic data, and the top-k parameter budget in SMG. AIM-Fair demonstrates that carefully curated synthetic data combined with selective parameter updates can substantially close demographic performance gaps while preserving overall model accuracy—opening a practical path toward fairer visual recognition systems that require no additional human annotation effort.

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